What KEV Misses, and What It Does to LEV: An Empirical Evaluation of NIST's Likely Exploited Vulnerabilities Metric

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Abstract

NIST's Cybersecurity White Paper 41 proposed Likely Exploited Vulnerabilities (LEV), a probability that a CVE has already been exploited, computed by compounding the CVE's EPSS history. It suggested using LEV to find exploited CVEs missing from CISA's Known Exploited Vulnerabilities (KEV) catalogue. We report what we believe is the first empirical evaluation of LEV.

Against KEV, LEV over-predicts by 17.3x on 105,250 CVEs. Two effects in its inputs inflate it. The EPSS v4 release rescaled scores overnight, accounting for 27-29% of LEV's mass. And LEV keeps growing, because EPSS rises rather than decays for CVEs no list records.

To separate LEV's error from KEV's incompleteness, we audit four exploitation lists for dependence. Two silently contain KEV, and a third contains a fourth. We then apply capture-recapture estimation, with explicit rules for when an estimate may be reported. KEV records at most 32% of the exploited CVEs that three evidence-citing lists observe. Where the estimate is identified, for LEV of 0.9 or above, LEV averages 0.985 but an estimated 48% of those CVEs were exploited. That is about 2x over-confident, not the 4.5x that KEV suggests. Lower strata are not identified.

A monotone correction calibrates LEV a year ahead but drifts with CVE age, and a learned aggregation of the same EPSS history does no better. Most new listings (59-75%) concern CVEs first scored within the previous 30 days, out of reach of any metric built on a CVE's history.

1 Introduction

Defenders cannot patch everything, so they rank. Two public resources dominate that ranking.

- The Exploit Prediction Scoring System (EPSS) gives each CVE a daily probability of exploitation in the next 30 days [2, 3].
- CISA's Known Exploited Vulnerabilities catalogue (KEV) lists CVEs with confirmed exploitation [4].

EPSS looks forward and KEV looks back. Neither says how likely it is that a given CVE has *already* been exploited without anyone recording it.

Mell and Spring proposed LEV to fill that gap [1]. LEV treats each 30-day EPSS score as the probability of exploitation in that window, and combines the windows since the CVE first appeared:

$$\mathrm{LEV}(v, d_0, d_n) = 1 - \prod_i \left(1 - \mathrm{epss}(v, d_i) \cdot w(d_i, d_n)\right),$$

where $d_i = d_0 + 30i$, and w is 1 for full windows and the remaining fraction for the last one. The white paper suggests two uses: flagging high-LEV CVEs absent from KEV as likely exploited, and measuring how complete KEV is. It also states that assessing LEV's performance requires collaboration with parties holding exploitation data [1]. We found no published evaluation (Section 8).

The obvious evaluation compares LEV with KEV, but it cannot separate two stories. If LEV says 0.99 and the CVE is not in KEV, either LEV is wrong or KEV missed it. The second story is the one the white paper proposes LEV for. A KEV-only evaluation therefore measures LEV's error and KEV's blindness together, in unknown proportion. That confound is not specific to LEV: any prioritization method validated against KEV inherits it [21-23].

We ask four questions:

- **RQ1.** Does a faithful reimplementation reproduce the white paper, and how calibrated is LEV against KEV?
- **RQ2.** How do LEV's inputs behave over time, and what does that do to LEV?
- **RQ3.** How much of the gap measured against KEV is KEV's incompleteness rather than LEV's error?
- **RQ4.** Can LEV be corrected, or used to anticipate the catalogues?

Our contributions:

1. **An evaluation of LEV and its inputs (RQ1, RQ2).** We reproduce the white paper's published values to within 11%. Against KEV, LEV over-predicts 17.3x while still ranking CVEs. We trace two input effects:
   - the EPSS v4 release rescaled scores in one day, and LEV compounds across it;
   - EPSS rises, rather than decays, for CVEs no list records, so LEV does not saturate.
2. **An estimate of catalogue incompleteness (RQ3).** We audit candidate exploitation lists for dependence and apply capture-recapture estimation with pre-declared identification rules. KEV records at most a third of what other evidence-citing lists observe. Where the estimate is identified, LEV is about 2x over-confident.
3. **What can and cannot be fixed (RQ4).** A monotone correction restores calibration a year ahead, but drifts because LEV's meaning changes with CVE age. A learned aggregation of the same EPSS history does no better. LEV gives a strong but low-recall early signal, and most listings happen too soon after publication for any history-based metric to anticipate.

All data snapshots, hashes and code are released, and one command regenerates every number in this paper (Section 11).

2 Background

2.1 EPSS

EPSS is a gradient-boosted model that scores every published CVE daily with the probability of exploitation activity in the following 30 days [2, 3]. Its ground-truth labels come from intrusion-detection and honeypot telemetry supplied by Fortinet, AlienVault OTX, the Shadowserver Foundation and GreyNoise [3]. The v3 model's 1,477 input features include:

- public exploit code (Exploit-DB, GitHub, Metasploit);
- offensive tool templates (Nuclei and others);
- membership of public exploitation lists, among them CISA KEV [3].

Two facts about this matter later. KEV is an *input* to EPSS, so a CVE's KEV listing can raise its later EPSS scores. And FIRST releases new model versions: v3 on 2023-03-07, v4 on 2025-03-17 and v5 on 2026-06-15 [44]. Each release can move every score in the corpus at once.

2.2 LEV

LEV is the equation in Section 1, applied from the CVE's first EPSS date d_0 to an analysis date d_n. Each 30-day window takes the EPSS score on its first day. The white paper also defines LEV2, which uses every daily score, each converted to a one-day probability. Because LEV is a product over windows, it cannot decrease as d_n advances. The white paper argues the growth should stall, because EPSS scores for old, unexploited CVEs decay toward zero [1].

2.3 KEV

CISA maintains KEV under Binding Operational Directive 22-01 [4]. An entry requires reliable evidence of exploitation, an assigned CVE and clear remediation guidance. `dateAdded` records when CISA catalogued the CVE, not when exploitation began. KEV is selective by design: it serves federal remediation priorities rather than aiming at complete coverage. Industry reports already show that it lists a minority of exploited CVEs [18, 41, 42]. What has been missing is an estimate built from open data, and an account of what that incompleteness does to evaluations that treat KEV as ground truth.

3 Data

EPSS. We use 67 monthly snapshots (the first of each month, plus the dates Table 5 of [1] anchors on) from 2021-04-14 to 2026-08-01: 14,456,555 scores for 367,311 CVEs. A further 185 daily files serve three checks. They locate model releases to the day, measure each release's one-day step, and bound the error from monthly sampling (Section 9).

CISA KEV. catalogVersion 2026.08.25, 1,676 entries. 1,665 have EPSS data; the other 11 were never scored and cannot receive a LEV value.

CVE List v5. Release `cve_2026-08-25_1700Z`, 192,054 records from 2022 onward, for CVSS, CWE and vendor.

Exploitation lists. The CIRCL Vulnerability-Lookup service publishes "exploited" sightings with the source of each [5]; we retrieved all 68,191 on 2026-09-28. VulnCheck's KEV [11] lists 5,371 records, each with the URL and date of every exploitation report behind it; we retrieved it with a community token on the same day. We also retrieved ENISA's EU Vulnerability Database known-exploited dump [6]. Section 4.6 describes how each became, or did not become, a list.

Analysis population. Unless stated otherwise, LEV is computed at d_n = 2026-08-01. The calibration cohort is the 105,250 CVEs first scored in the EPSS v3 era (from 2023-03-07), at least six months before d_n, so each had time to reach KEV.

We exclude CVEs that EPSS backfilled. When EPSS v2 launched on 2022-02-04, its coverage jumped from 80,800 to 168,325 CVEs, most of them old, and taking first-EPSS date as a publication proxy would count them as new. A CVE is kept only if EPSS first scored it within one calendar year of its CVE-ID year. 495 cohort CVEs are in KEV, a base rate of 0.47%.

Which LEV, which population. The paper uses two LEV variants and three populations. Table 0 lists every LEV-versus-KEV ratio that appears later, so the numbers can be compared.

population	N	LEV variant	LEV / KEV rate
calibration cohort (Sections 5.1-5.2)	105,250	as specified [1]	17.3x
calibration cohort	105,250	version-normalized (Section 4.3)	12.6x
capture-recapture population (Section 5.4)	54,612	as specified	11.2x
capture-recapture population	54,612	version-normalized	7.9x
prospective test set (Section 5.5)	44,316	version-normalized	6.3x

Table 0: Every LEV-versus-KEV ratio in the paper. The capture-recapture population is younger (first scored 2024-10-24 to 2026-01-30), so its LEV is lower. Phase 2 analyses use version-normalized LEV; Section 5.4 also gives the as-specified values.

Provenance. Every fixed input is recorded with its SHA-256 and verified on load. KEV is vendored, and the CVE List is pinned to its release tag. EPSS files are checked against a manifest. Sources served from moving branches or live APIs (Exploit-DB, Metasploit, CIRCL, EUVD) are snapshotted and committed with the code rather than re-fetched.

4 Methods

4.1 Reimplementation and reproduction

We implement LEV and LEV2 as specified, including the white paper's inclusive day count for the final partial window and LEV2's division of each daily score by 30. When a window starts on a date we do not hold, we take the next available snapshot, the rule the white paper gives for its own code [1, Sec. 10.3].

We compare the average LEV of CVEs published since each date in Table 5 of [1] with the published values, at d_n = 2025-03-01. That is close to the white paper's own analysis date, which it does not state. LEV grows with d_n, so the date matters. We therefore also report reproduction error at every monthly d_n from 2024-09-01 to 2025-09-01, with and without the backfill filter of Section 3.

4.2 Calibration

We bin the cohort into ten LEV buckets and compare each bucket's mean LEV with the fraction of its CVEs in KEV, with Wilson score intervals [7] for the observed rates.

- We test for a monotone trend across buckets with the Cochran-Armitage test.
- We report the Brier score [9] and expected calibration error (ECE) over the same ten bins [10].
- We repeat the comparison by CVE age at d_n (6-12, 12-24 and 24+ months), and for maturity rules of 3, 6 and 12 months.

4.3 Version normalization

EPSS releases can move every score at once, and LEV multiplies scores across them. We estimate how much of LEV this contributes in two ways.

1. **Percentile mapping.** We map every score produced after the v4 release back onto the v3 scale. Each score keeps its percentile within its own day's file, and we substitute the score at that percentile in the last v3 file (2025-03-16).
2. **Step-only.** At each release (v4 on 2025-03-17, v5 on 2026-06-15) we fit the map once, from the last file before to the first file after, and then hold it fixed. This removes the one-day rescaling and leaves organic drift within a version untouched.

Neither method can separate a corpus-wide shift from genuine per-CVE improvement in the newer model. The removed mass is therefore an estimate of the artifact, not a corrected value. Phase 2 analyses use percentile-mapped ("version-normalized") LEV throughout; Section 5.4 also reports as-specified LEV.

4.4 Analysis-date sweep

We take two fixed cohorts:

- 13,685 CVEs first scored March-September 2023;
- 7,558 CVEs first scored May-September 2021.

For each, we compute LEV at every month from 2023-10-01 to 2026-08-01. We compare linear, logarithmic, two- and three-parameter saturating, quadratic, exponential and power-law growth. Monthly residuals are strongly autocorrelated (lag-1 correlation 0.6-0.9), so we judge the comparison with autocorrelation-robust (HAC) standard errors and a moving-block bootstrap, not AIC alone. The test of the white paper's premise is direct: we track mean EPSS for cohort CVEs that no exploitation list ever records.

4.5 Circularity and early signal

Because KEV is an EPSS input, we test how much listing moves scores, with two designs that cannot select on the outcome.

The jump at listing. For each KEV CVE we compare EPSS in the last snapshot before `dateAdded` with the first snapshot on or after it. We net this against non-KEV CVEs in the same starting-score decile over the same pair of snapshots, and exclude pairs that straddle a model release.

Pre-listing LEV. We compute each KEV member's LEV using only snapshots dated strictly before its listing. We compare it with non-KEV CVEs first scored on the same date and evaluated on the same date. As a forward check, we ask how well LEV of 0.5 or more at a fixed CVE age (three or six months) predicts later listing.

4.6 Capture-recapture

Lists. An exploitation list is a set of CVEs that a source records as exploited in the wild. We use four:

- CISA: CISA KEV.
- SENSOR: CIRCL sightings from Shadowserver's honeypot feed (969 CVEs), which records attempted exploitation directly.
- VULNCHK: VulnCheck's KEV, rebuilt from its per-report evidence (4,815 CVEs).
- SOCIAL: CIRCL sightings from Telegram and similar channels (2,396 CVEs).

Dependence audit. We audited every candidate for dependence before use. A list that contains another makes their overlap equal to the smaller one by construction. Three such cases turned up:

- CIRCL's aggregated KEV catalogue contains all 1,676 CISA KEV CVEs, so we exclude it.
- VulnCheck's KEV includes CISA KEV by its own documentation. We rebuild it from its reports and drop every report hosted on cisa.gov.
- The third was less obvious. VulnCheck's largest single source is Shadowserver's dashboard (401,967 reports), and every one of the 969 SENSOR CVEs appears in VulnCheck's Shadowserver subset. We drop those reports too.

What remains is VulnCheck's own honeypot canaries, SANS DShield, CrowdSec, vendor advisories and threat reports. For 78% of CISA-listed CVEs, the earliest remaining report predates CISA's listing (median 28 days), so the remainder is not an echo of CISA.

EU KEV lists 66 CVEs, and 55 are also on CISA KEV. 36 of the 41 in our KEV snapshot (88%) carry CISA's own date, so we treat it as derivative. ENISA's exploited-flag search returns the same consolidated set. Exploit-DB records exploit availability, not exploitation, and is excluded.

Primary lists. Telegram channels carry mass lists of CVEs that actors claim to use, and social signals are known to be noisy [34]. We therefore take CISA, SENSOR and VULNCHK, the three whose entries each cite evidence, as primary. We report every other configuration alongside.

Population. SENSOR begins on 2024-10-24. We restrict to the 54,612 CVEs first scored between 2024-10-24 and 2026-01-30, so every list could observe every CVE from its first score, and apply the six-month maturity rule. 734 of these appear on at least one primary list. We stratify by version-normalized LEV into [0, 0.1), [0.1, 0.9) and [0.9, 1], to absorb capture heterogeneity.

Models. Two-list estimates among all four lists disagree by 2.4x (530 to 1,256), so the lists are not independent and dependence has to be modelled.

- We fit Poisson log-linear models to the seven observable capture patterns, under every hierarchical structure of two-way interactions.
- We assume no three-way interaction, the standard identifying assumption [12, 13].
- We select by AIC and bootstrap a 95% interval from the fitted cell means (1,000 replicates).
- A model whose interaction is fitted to an empty cell reaches its maximum likelihood only at the boundary. We score it by its extended MLE and count its residual degrees of freedom honestly [45]. A model left with none is excluded as saturated.

Chao's estimator [14] is a lower bound that stays valid under heterogeneous capture probabilities. We use its form for t lists, $S + \frac{t-1}{t} f_1^2/(2f_2)$, with a log-normal interval.

When not to report an estimate. Log-linear estimates of the unseen cell can be driven entirely by the choice of interaction terms, especially when lists do not overlap [36]. We declare a population identified only if *every* model within 2 AIC of the best:

- stays within the population;
- does not fall below the lower end of Chao's interval;
- and, for a pooled population, does not fall below the sum of the per-stratum Chao bounds;

and if those models agree within a factor of 2 [15]. Otherwise we report no point estimate.

The truth ladder. Rather than one estimate, we report LEV's over-confidence (mean LEV over the exploitation rate) against successively stronger notions of truth, each adding one assumption: KEV alone; KEV or SENSOR; the union of the primary lists; Chao's bound; the log-linear estimate.

Every rung is a lower bound on, or an estimate of, the number of CVEs exploited, so every over-confidence figure is an upper bound on LEV's error. A reader can stop at whichever assumption they accept.

4.7 Correction curve

We fit a monotone map from LEV to observed exploitation by isotonic regression [16], by Platt scaling [17], and by a Platt variant that also takes log CVE age. There are two targets: KEV on the calibration cohort, and the union of the primary lists on the capture-recapture population. All scores are on held-out CVEs, under three schemes:

- five-fold cross-validation, the in-distribution case;
- a split by first-score date, which trains on older CVEs and tests on newer ones;
- the deciding scheme, which is prospective. The map is fitted at d_n = 2025-08-01 with outcomes known then, and applied at 2026-08-01 to the 44,316 CVEs first scored after the training cohort, none of which it saw.

Prospective ratios carry bootstrap intervals over test CVEs. Appendix A tests whether a learned aggregation of the same EPSS history beats LEV.

5 Results

5.1 RQ1: the reimplementation reproduces the white paper

CVEs published since	CSWP 41	ours	difference
2023-03-07	0.015	0.0162	+8%
2022-02-04	0.047	0.0480	+2%
2021-04-14	0.087	0.0839	-4%
all CVEs	0.146	0.1299	-11%

Table 1: Average LEV at d_n = 2025-03-01 against Table 5 of [1].

All four values fall within 11%. The fourth row matches only with the backfill filter of Section 3. We added that filter after the row first missed by 75%, but it is justified independently: first-EPSS date is a publication proxy only for CVEs EPSS did not backfill.

The match does not depend on picking one date. With the filter, every row is within 12% at each monthly d_n from 2024-09-01 to 2025-03-01 (mean error 4-6%). Error then rises from April 2025, after the v4 release, to 70% by September. Without the filter, no date brings the rows within 30%. This validates the implementation. It does not validate the metric.

5.2 RQ1: against KEV, LEV over-predicts 17.3x but ranks

Across the 105,250-CVE cohort, mean LEV is 0.081 against a KEV rate of 0.0047: LEV over-predicts by 17.3x. The 95% interval is 15.8-18.9x, from Poisson error on the 495 KEV CVEs; Brier 0.0373, ECE 0.0765.

The observed rate climbs from 0.03% in the lowest LEV bucket to 12.9% in the highest, where LEV averages 0.987 (Figure 1). LEV ranks across its whole range: even within the eight middle buckets the trend is strong (Cochran-Armitage z = 9.2).

Over-prediction grows with CVE age (Table 1a), the first sign of an effect that recurs in Sections 5.3 and 5.5. The six-month maturity rule is not driving the result: with 3 or 12 months the ratio is 16.0x or 19.1x.

CVE age at d_n	N	KEV CVEs	LEV / KEV rate	KEV rate at LEV ≥ 0.9
6-12 months	22,043	82	6.6x [5.3, 8.3]	27.1%
12-24 months	40,875	200	15.1x [13.1, 17.4]	16.3%
24+ months	42,332	213	23.4x [20.5, 26.9]	9.8%

Table 1a: Calibration against KEV by CVE age.

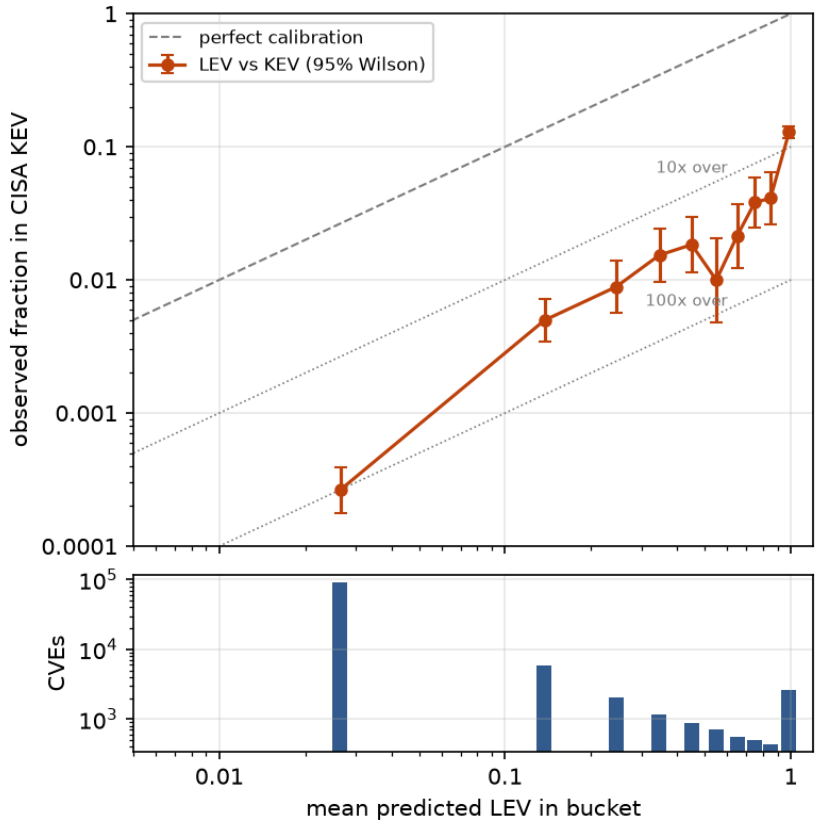


Figure 1. Reliability of LEV against CISA KEV, log scales. Error bars are 95% Wilson intervals; dotted lines mark 10x and 100x over-prediction.

5.3 RQ2: EPSS releases put an artifact into LEV

EPSS v4 went live on 2025-03-17 [44]. Its version string, v2025.03.14, is a build date, so bracketing by version string places the change two weeks early and measures almost no step.

On an unchanged set of 13,552 cohort CVEs, mean EPSS goes from 0.00862 on 2025-03-16 to 0.02640 the next day, a 3.06x step (Figure 2). The whole corpus moves only 1.21x, and the cohort's rank correlation across the step is 0.61. For young CVEs, v4 re-ranked as much as it rescaled. Of the rise in cohort EPSS over 2023-2026, 64% (v3 cohort) and 70% (v1/v2 cohort) happens at release boundaries rather than between them.

EPSS v5 (2026-06-15) moved the same cohort the other way: its mean fell to 0.79x of the previous day's.

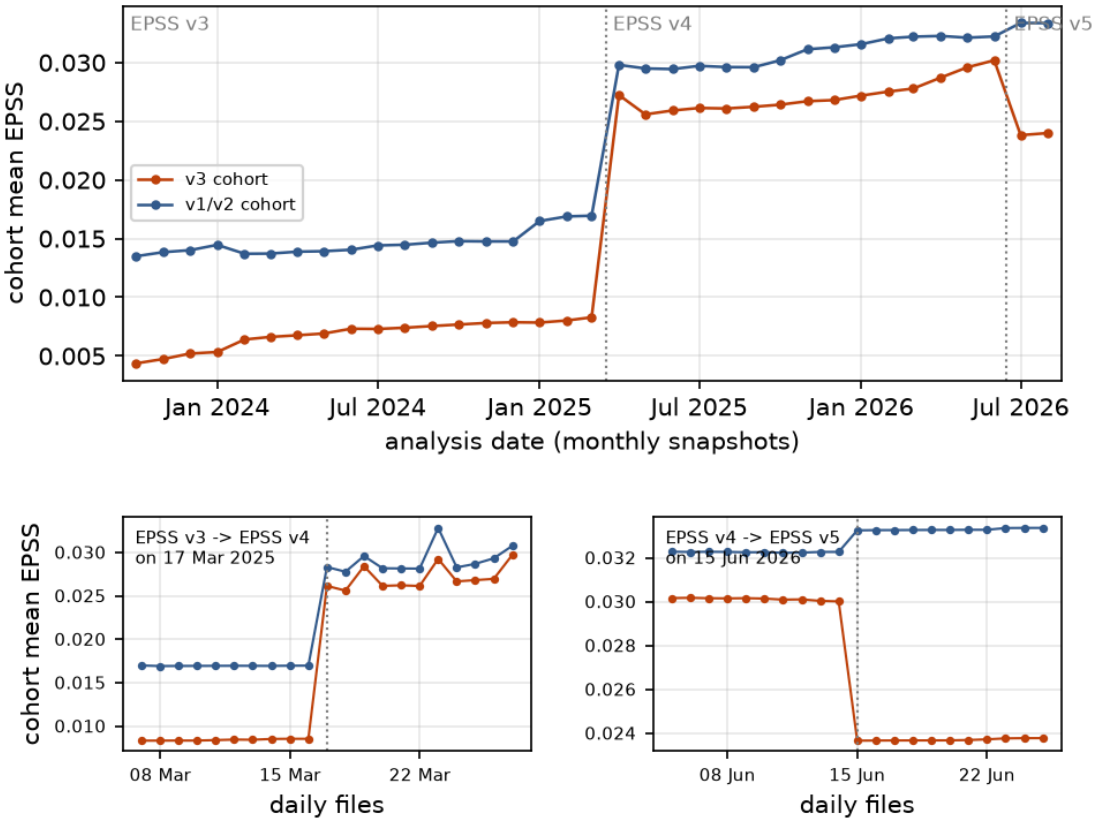


Figure 2. Cohort mean EPSS, monthly, with daily detail around the v4 and v5 releases.

LEV multiplies scores across these boundaries as if they belonged to one measurement series. Removing the one-day rescaling at the v4 and v5 boundaries lowers mean LEV by 27-29%:

- 26.9% by percentile mapping;
- 29.0% step-only.

Over-prediction against KEV falls from 17.3x to 12.3-12.6x (normalized: Brier 0.0289, ECE 0.0547). Treating *all* of the cohort's re-ranking at v4 as artifact gives an upper bound of 62%.

Version mixing therefore explains between a quarter and three-fifths of LEV's mass. Even the upper bound leaves LEV over-predicting KEV by 6.6x.

5.4 RQ2: LEV grows without saturating, because EPSS does not decay

For the 2023 cohort, mean LEV rises from 0.008 to 0.135 between October 2023 and August 2026 (Figure 3). Monotone growth is a property of the equation, not a finding. The empirical questions are whether it slows, and whether the premise behind an expected slowdown holds.

It does not slow. A three-parameter saturating model collapses to its linear limit. Within the retraining-free stretch, the curvature term is significant under autocorrelation-robust errors (HAC z = 3.33). A moving-block bootstrap rejects linear (p = 0.034) and saturating (p = 0.018) growth in favour of accelerating growth. The data cannot separate a quadratic from a power law.

The premise fails directly. For the 13,390 cohort CVEs that no exploitation list records, mean EPSS rises from 0.0022 to 0.0046 within the single v3 model (p = 3 x 10^-8), and reaches 0.0170 by August 2026 across releases. Scores for old, unlisted CVEs rise; they do not decay. With no decay, LEV keeps compounding and, over time, more of the corpus approaches 1.

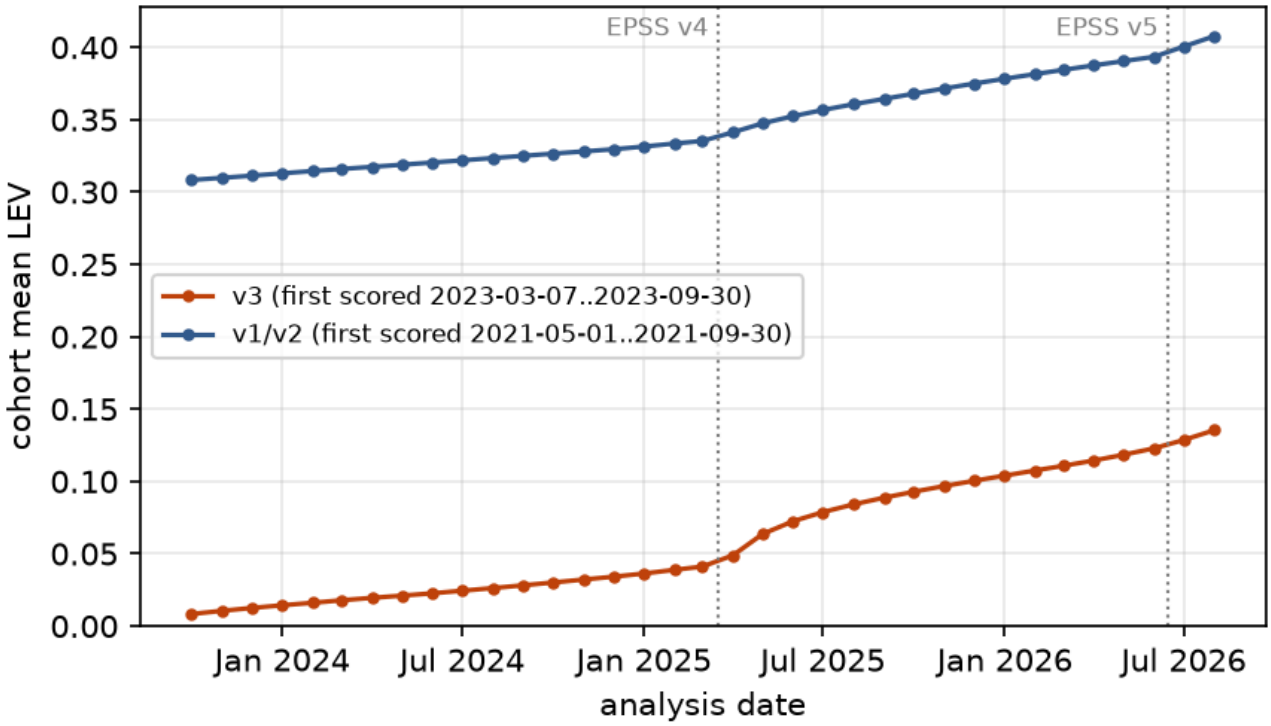


Figure 3. LEV for fixed cohorts as the analysis date advances. Dotted lines mark EPSS releases. Further panels are in Appendix B.

5.5 RQ4: LEV's early signal is strong but rare, and its top bucket is partly circular

The listing feedback. At the listing date, the median KEV CVE's EPSS does not move relative to matched non-KEV controls. But a quarter of listed CVEs rise at least 2.85x more than their controls. The KEV input feature matters for a sizeable minority of listings rather than all of them.

Before listing. 56% of cohort KEV members were listed in the same month EPSS first scored them, so no history-based metric could have flagged them first. 220 have a pre-listing history (median 2.8 months).

- **Before listing, LEV was 25 times more likely to be high.** Using only snapshots dated strictly before listing, 15% of the 220 had LEV of 0.5 or more, against 0.58% of non-KEV CVEs first scored on the same date and evaluated on the same date.
- **The typical member already stood out.** Its pre-listing LEV (median 0.005) sat at the 92nd percentile of its matched controls.
- **After listing, the median is 1.0.** Over the same period, same-age controls move only from 0.0013 to 0.017, so elapsed time explains little of that jump.

As a forecaster. At a fixed CVE age of three months, LEV of 0.5 or more picks out CVEs listed later with 6.0% precision (59 times the base rate), but with 10% recall.

So LEV carries a real early signal, but for few CVEs. Most of the top bucket's KEV concentration (Section 5.2) arrives after listing, and is partly circular.

5.6 RQ3: how much of the gap is the catalogue's

Table 2 gives the truth ladder on the 54,612-CVE capture-recapture population with the three primary lists.

truth	assumption added	exploited	over-confidence	as-specified LEV
CISA KEV	KEV is complete	236	7.93x [6.98, 9.05]	11.16x
CISA or SENSOR	honeypot sightings are real	336	5.57x [5.01, 6.22]	7.84x
union of primary lists	every listed report is real	734	2.55x [2.37, 2.75]	3.59x
Chao lower bound	closed population	974	1.92x [1.79, 2.04]	2.70x
log-linear	no three-way interaction	not identified		

Table 2: LEV over-confidence against successively stronger notions of truth, all LEV strata pooled, version-normalized LEV, with as-specified LEV alongside. Intervals are 95%: Poisson on the counts, log-normal for Chao.

By counting alone, over-prediction falls from 7.9x to 2.6x. 68% [64, 71] of the CVEs that any primary list records are absent from KEV: 60% [55, 64] if VulnCheck entries resting on a single report are dropped. KEV recorded at most 32% of the exploited CVEs the primary lists observed, and at most 24% under Chao's bound.

The pooled log-linear estimate falls below Chao's bound and is not reported. Table 3 stratifies by LEV.

LEV	N	mean LEV	vs KEV	union	Chao	log-linear
[0.0, 0.1)	52,012	0.012	21.2x	2.36x	1.05x [0.83, 1.29]	not identified
[0.1, 0.9)	1,964	0.322	9.16x	3.21x	2.54x [2.24, 2.76]	not identified
[0.9, 1.0]	636	0.985	4.54x	2.26x	2.05x [1.93, 2.14]	**2.04x [1.81, 2.32]**

Table 3: Over-confidence by LEV stratum, primary lists, version-normalized LEV.

The top stratum is identified. LEV averages 0.985, and we estimate that 48% [42%, 54%] of those CVEs were exploited: about 2x over-confident, against the 4.5x that KEV alone suggests. As-specified LEV gives the same figure there (2.04x).

The middle stratum is not identified. No CVE in it is on both CISA and SENSOR. The two best models fit that empty cell at the boundary and have no residual degrees of freedom, and the model that remains falls below Chao's bound. Chao's bound alone caps over-confidence there at 2.5x (3.2x by counting).

The lowest stratum is not identified either, and it holds 95% of the CVEs. Chao's bound alone brings it to 1.05x [0.83, 1.29]. There, LEV may not over-predict at all.

Does this hinge on the choice of lists? Table 4 repeats the estimate under every configuration.

lists	LEV [0.9, 1.0]	LEV [0.1, 0.9)	all LEV
CISA, SENSOR, VULNCHK (primary)	2.04x [1.81, 2.32]	not identified	not identified
CISA, SENSOR, strict VULNCHK	2.12x [1.85, 2.43]	not identified	2.64x [2.34, 2.93]
CISA, SENSOR, SOCIAL	2.07x [1.76, 2.45]	not identified	not claimed (0.99x)
CISA, SOCIAL, VULNCHK (no SENSOR)	1.92x [1.66, 2.23]	1.22x [0.69, 1.70]	not identified
all four	not identified	not identified	not claimed (0.35x)

Table 4: Log-linear over-confidence by list configuration. “Not claimed”: passes the rules pooled, but pooled over strata that the stratified analysis rejects.

Wherever the top stratum is identified, its estimate lies between 1.92x and 2.12x. That includes the strict configuration, which keeps a VulnCheck CVE only if a sensor saw it or at least two independent sources reported it (2,912 CVEs). That check matters: VulnCheck supplies most CVEs seen by a single list, and 65% of VulnCheck-only CVEs rest on one text report.

Three further checks:

- **No SENSOR.** Shadowserver is one of EPSS’s label sources, so EPSS is trained to predict roughly what SENSOR records. Counting with CISA and VULNCHK alone gives 2.65x pooled and 2.38x at the top.
- **S×V dependence.** Removing Shadowserver-sourced reports from VulnCheck may create negative dependence between SENSOR and VULNCHK by construction. Forcing that interaction into every model leaves nothing identified; at the top it gives 2.24x.
- **The listing boost.** The boost of Section 5.5 lifts some KEV members into the top stratum. They are true positives, so removing the boost would, if anything, raise the top stratum’s 2x.

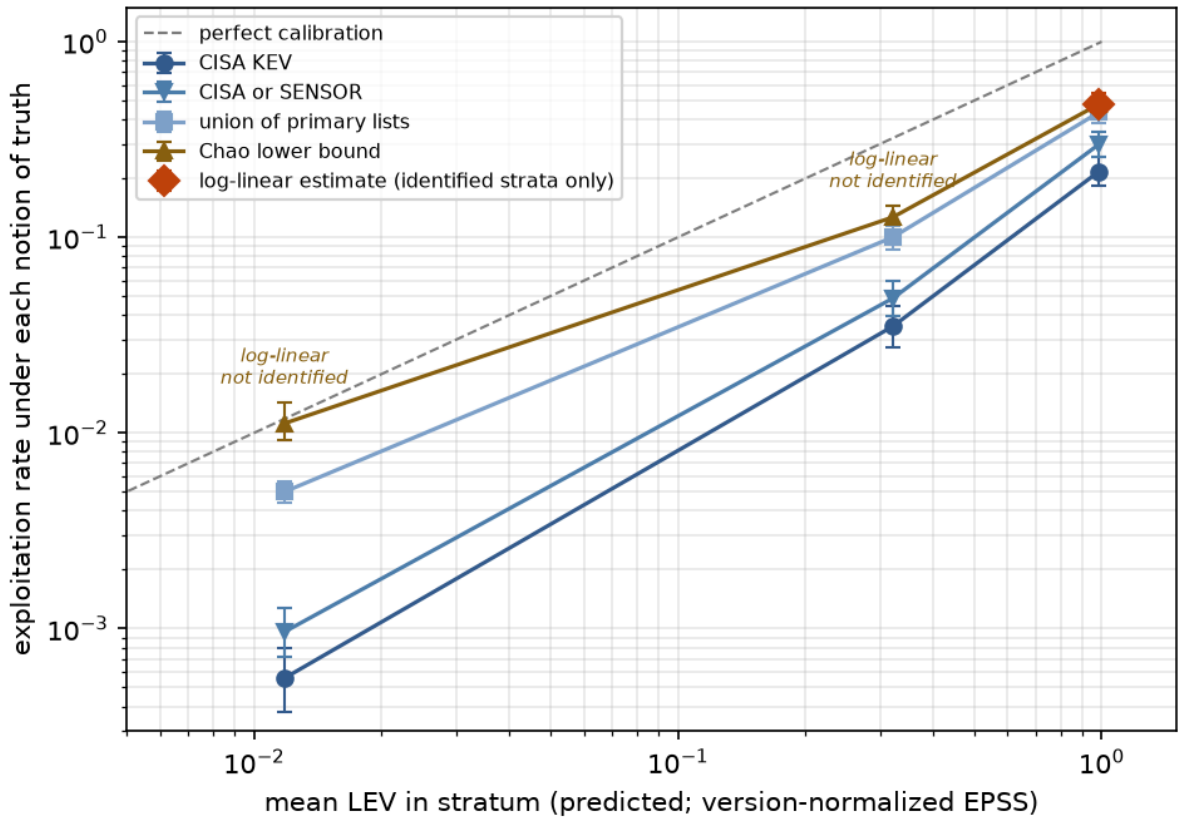


Figure 4. LEV against progressively broader ground truth, by LEV stratum, primary lists. Error bars are 95% intervals.

5.7 RQ4: a correction curve, and why it drifts

method	Brier	ECE	predicted / observed
LEV (version-normalized)	0.0113	0.0230	6.30x [5.56, 7.27]
LEV (as specified)	0.0136	0.0339	8.81x [7.77, 10.18]
isotonic	0.0036	0.0020	0.61x [0.53, 0.70]
Platt	0.0037	0.0015	0.66x [0.58, 0.76]
Platt with CVE age	0.0036	0.0014	0.72x [0.63, 0.83]

Table 5: Prospective validation against KEV. Fitted at 2025-08-01; scored at 2026-08-01 on 44,316 CVEs the fit never saw.

Under cross-validation, any of the three maps restores calibration (predicted over observed 1.00-1.01x). Prospectively against KEV, each map cuts calibration error 11-17x, but all under-predict by 28-39% a year later.

Against the union of the primary lists, the maps fitted on 111 training positives and scored on 597 test positives give:

- isotonic: 1.00x [0.93, 1.08];
- Platt: 0.80x [0.74, 0.87];
- age-aware Platt: 0.88x [0.81, 0.95].

Much of the KEV-target drift therefore reflects KEV's narrow and lagging coverage rather than the correction itself.

Transfer across birth cohorts is worse. A map fitted on CVEs first scored before 2025-06-01 under-predicts the younger ones at the same analysis date (isotonic 0.30x, Platt 0.52x against the union), and the age-aware Platt model overshoots (2.32x).

The drift has a cause we can name. An old CVE reaches a given LEV through many windows of modest EPSS, and a new one through a few high scores. The second is stronger evidence, so the same LEV value carries a different probability at different ages. That is Table 1a seen from another side. A Platt model that also takes log age recovers about a quarter of the drift.

A learned aggregation does no better (Appendix A). We also gave a logistic regression and a gradient-boosted model the same EPSS history LEV sees, and nothing else. Neither ranks exploited CVEs better than LEV or current EPSS a year ahead, on any target. In a forward test across three origins, no method anticipates new list entries better than current EPSS. 59-75% of each window's listings concern CVEs first scored less than 30 days earlier, which no history-based metric can see in advance.

An apparent learned gain appears only with un-normalized inputs and KEV-listed CVEs in the target, and vanishes when either is removed.

LEV	corrected to KEV	corrected to union of primary lists
0.10	0.8% [0.5, 1.0]	6.1% [4.6, 7.5]
0.50	2.7% [1.9, 4.1]	15.5% [11.5, 19.0]
0.90	6.7% [4.4, 8.4]	20.2% [16.5, 29.5]
0.99	8.2% [6.4, 10.1]	32.5% [26.9, 37.4]
1.00	27.8% [24.3, 30.9]	76.3% [68.0, 85.9]

Table 6: The isotonic correction, fitted on all data, with 95% bootstrap intervals (400 resamples of CVEs). Both targets are catalogue appearance, so both are lower bounds on true exploitation. The intervals are several points wide: the curve gives the scale of LEV's error, not a precise mapping for a single score.

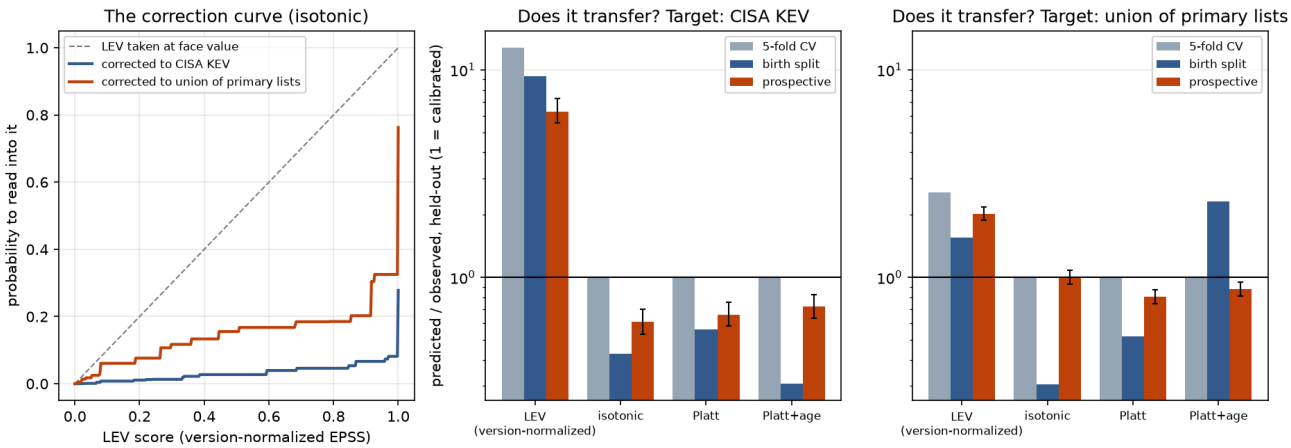


Figure 5. Left: the correction curves. Centre and right: held-out predicted/observed under each validation scheme, against KEV and against the union of primary lists.

6 Discussion

What LEV is good for. LEV ranks. Its extremes separate (the top bucket's KEV rate is nearly 500 times the bottom bucket's), and the trend holds across the middle of its range. Its values should not be read as probabilities. A LEV of 0.9 corresponds to about 7% chance of KEV listing, and about

20% chance of appearing on any primary list.

What LEV is not good for yet. The white paper proposes LEV for finding exploited CVEs that KEV misses. Our results cut both ways.

- LEV's apparent over-confidence is substantially KEV's incompleteness, which is the premise of the proposal.
- Before listing, a high LEV is 25 times more common among CVEs that KEV later lists.
- But it flags only 15% of them, and most listings happen within a month of a CVE's first EPSS score, before any history accumulates.

LEV can support the proposal at the margin. It cannot carry it.

What would make LEV more reliable. Five changes follow from the results.

1. Compute LEV within one EPSS model version, or normalize across releases, since a release otherwise enters as evidence.
2. State the analysis date with every published LEV value, since values are not comparable across dates.
3. Stop accumulating, or decay, windows for CVEs with no recent evidence, or accept that LEV drifts toward 1.
4. Read LEV through a correction that is refitted as CVEs age, not at face value.
5. Validate against more than KEV: KEV is both an EPSS input and incomplete.

KEV coverage, and evaluations built on it. Empirical Security reports that KEV covers at most about 8.7% of CVEs with observed exploitation activity, from proprietary telemetry [18]. Our figures come from open data on a younger population: KEV recorded at most 32% of what the primary lists observed, and at most 24% under Chao's bound.

The direction agrees. The levels differ as they should, since three lists see less than one vendor's full telemetry. Any method tuned or tested on KEV [21-23] measures, in part, CISA's visibility. The learned-model reversal in Appendix A shows how such an evaluation can report gains that are pure feedback.

Two definitions of "exploited." Part of the KEV gap is definitional. EPSS is trained on intrusion-detection and honeypot detections, so LEV estimates the probability of observed exploitation *activity*. KEV lists confirmed, actionable exploitation. Against sources closer to EPSS's own definition, LEV in its top stratum looks about twice too confident rather than four or five times.

7 Threats to validity

We record 55 threats in the released repository (`docs/THREATS.md`). The ones that bear most on the conclusions:

- **Version-normalized LEV.** Phase 2 uses version-normalized LEV. As-specified LEV gives higher over-confidence everywhere except the top stratum, where the two agree (Table 2, Section 5.6).
- **Lists sample different things.** KEV only lists CVEs with remediation guidance and federal relevance. Honeypots see only internet-reachable exploitation. These structural zeros violate the shared-population assumption, and stratifying by LEV does not repair them; they bias capture-recapture estimates downward.

- **Catchability changes with age.** In CIRCL's comparison, Shadowserver's catalogue lists CVEs at a median age of 532 days, against 252 for CISA [40]. Our population is observed for only 6-21 months, so late listings are still to come.
- **Strata depend on capture.** KEV listing raises later EPSS (Section 5.5), so capture moves CVEs into the top stratum. Stratifying on pre-listing LEV would avoid this.
- **Monthly sampling.** LEV's windows are 30 days and our snapshots are about 30.4 days apart. On 481 CVEs stratified by LEV decile, LEV from true daily windows differs from ours by 3.9% of mean LEV (population-weighted), with no consistent sign.
- **Publication-date proxy.** We use first EPSS appearance as publication date, with a backfill filter added after a first miss (Section 5.1).
- **Circularity.** KEV is an EPSS input, and Shadowserver is an EPSS label source. We measure the first and drop the second as a check; we cannot remove either from EPSS.
- **List dependence and sparse cells.** The log-linear model assumes no three-way interaction. We report estimates only where identified, and Table 4 shows how each configuration changes them.
- **VulnCheck precision and dates.** 65% of VulnCheck-only CVEs rest on one text report; the strict configuration keeps the top-stratum result. VulnCheck dates come from a later snapshot and may be backdated, which affects the forward test of Appendix A. That test is also run on CISA and SENSOR dates alone.
- **Nested sources.** Two catalogues contain CISA KEV, and VulnCheck contains Shadowserver's list. Each was removed by source before modelling. A containment we did not detect would bias the estimates downward.
- **False sightings.** Telegram sightings include claims we cannot verify. Every headline figure is also given with that list excluded.
- **A short, young population.** SENSOR starts in October 2024, so the capture-recapture population covers CVEs first scored over 15 months.
- **Normalization.** Neither normalization separates corpus-wide shift from genuine improvement in the newer model, so the artifact is reported as a range.
- **KEV dates.** `dateAdded` is a cataloguing date, so timing comparisons are lagged by an unknown amount.
- **Search scope.** Our search for prior work covered 30 queries across arXiv and the web, plus the venues in Section 8. It was not a systematic review.

8 Related work

Exploit prediction. Learning which vulnerabilities get exploited predates EPSS:

- case-control comparisons of severity scores against exploitation [33];
- social-media signals [34];
- exploit-development forecasting [30];
- exploit prediction for remediation policy [32].

EPSS was introduced by Jacobs et al. [2] and extended in version 3 [3]. Parla evaluates EPSS on high-severity CVEs in KEV [19]. Koscinski et al. document how often vulnerability scores disagree [35]. Nucleus Security reports that EPSS moves far more after KEV listing than before [43], consistent

with Section 5.5. None of these evaluates LEV.

LEV. Beyond the white paper [1], an independent reimplementation by the Risk Based Prioritization project questions LEV's small-probability approximation and its independence assumption [20]. It does not validate LEV against exploitation data. LEV is already consumed downstream: Huang et al. use it, alongside EPSS and KEV, as an exposure probability for incident mitigation in cyber-physical systems [26]. We found no empirical evaluation of LEV.

KEV as ground truth. Recent frameworks combine CVSS, EPSS and KEV [21], build composite risk indicators evaluated on KEV [22], or train a ranking model with KEV entries as weak supervision [23]. Shaffer and Voicu model exploitation hazard with a decay calibrated on KEV timing [24]; our cohorts show EPSS rising, not decaying.

Industry analyses already show that KEV is incomplete and lagging [18, 41, 42], and CIRCL has compared five KEV-style catalogues descriptively [40]. We add a population estimate, the dependence audit those comparisons need, and its consequence for one metric.

Learning from incomplete labels. Exploitation lists give positives and unlabelled CVEs, the setting of positive-unlabeled learning [27, 31]. Under its selected-completely-at-random assumption, ranking evaluations survive incompleteness. Our vendor audit shows that assumption fails for KEV. We follow the temporal-split and label-quality guidance of the security-ML pitfalls literature [28, 29].

Multiple-systems estimation. Capture-recapture over administrative lists is standard in epidemiology [12]. It has been used to estimate casualties and modern slavery, where its sensitivity to list dependence and sparse, non-overlapping cells is well documented [25, 36]. In software engineering it estimates defects missed by inspections [39]. Overlap between independent discoveries has been used to estimate vulnerability rediscovery rates [37, 38]. We found no prior application to vulnerability exploitation.

9 Conclusion

Evaluated against KEV, LEV looks badly over-confident. Much of that is not LEV's doing.

- EPSS releases put an artifact into the product LEV computes.
- EPSS does not decay for unlisted CVEs, so LEV accumulates without the saturation its design assumes.
- KEV records a minority of the exploitation that other evidence-citing sources observe.

Once that is accounted for, LEV's most confident scores over-predict estimated true exploitation by about 2x, rather than the 4.5x that KEV suggests. Below that, the data bound the error but do not identify it: at most 2.5x in the middle stratum, and possibly none in the lowest.

A monotone correction brings LEV's values in line with the broader ground truth a year ahead, though it must be refitted as CVEs age. Neither LEV nor a learned aggregation of EPSS history can anticipate most listings.

The lowest LEV stratum holds most CVEs, and four lists cannot identify it. A fifth independent list with sensor coverage, such as GreyNoise's or Shadowserver's own KEV catalogue [40], is the natural next step. The method is not specific to LEV: any metric validated against KEV can be re-scored against the same truth ladder.

10 Ethics and data

All inputs are public or available under a free community licence. No personal data is processed.

- CISA KEV is a US-government work and is vendored with the code.
- EPSS scores, the CVE List and CIRCL sightings are used under their providers' terms.
- VulnCheck's terms do not say whether its data may be redistributed, so its snapshot is not released. Reproducing the four-list results needs a free community token, and all other results reproduce without it.
- Telegram-derived sightings are used only as aggregated counts per CVE, as CIRCL publishes them.

11 Data and code availability

[Repository URL]. `make fetch` downloads the pinned inputs, `make verify-data` checks them against recorded hashes, and `make reproduce` regenerates every table and figure offline. SHA-256 hashes, release tags and retrieval dates for all inputs are listed in `DATA_SNAPSHOT.md`.

Appendix A: a learned aggregation of EPSS history

Question. LEV combines a CVE's EPSS history with one fixed rule. Would a model that sees the same history, and nothing else, combine it better? We restrict the inputs so that the comparison tests LEV's rule, not the value of extra data.

Design. Seven features, all computed from version-normalized EPSS dated on or before the analysis date: LEV, log CVE age, and current, peak, mean, first and three-month-trend EPSS. There are two learners:

- L2 logistic regression, with probability features on the logit scale;
- gradient-boosted trees, monotone in LEV and heavily regularized.

The choice between them is fixed by a rule that sees only training data: fit on the older two-thirds of the training cohort, validate on the newer third, keep the higher average precision (AP). We wrote this rule after a first, unreported run, and we report both models throughout. Baselines are LEV, current EPSS, peak EPSS, and the maps of Section 4.7.

Experiment 1 repeats the prospective test of Section 4.7 with three targets:

- the union of primary lists;
- CISA KEV;
- the union with every KEV-listed CVE removed from training and test, which cuts the path by which a listing raises later scores.

The logistic model's bootstrap resamples training *and* test CVEs and refits, so its intervals include training variance.

Experiment 2 is forward. At origin t we take every CVE that is at least 30 days old and on no list. We compute features at t and label the CVE positive if it is first listed in the next 182 days. Training uses the origin one horizon earlier. There are three test origins (2025-08-02, 2025-10-31, 2026-01-30), and a second target that uses only CISA and SENSOR dates.

method	union AP (pred/obs)	KEV AP (pred/obs)	union without KEV AP (pred/obs)
LEV	0.399 (2.03x)	0.324 (6.30x)	0.228 (2.66x)
current EPSS	0.404 (0.65x)	0.457 (2.02x)	0.221 (0.72x)
peak EPSS	0.424 (0.97x)	0.459 (3.03x)	0.237 (1.17x)
isotonic	0.337 (1.00x)	0.239 (1.29x)	0.203 (0.94x)
Platt with age	0.366 (0.88x)	0.329 (0.61x)	0.180 (1.20x)
logistic	0.398 (0.99x)	0.196 (1.52x)	0.190 (0.74x)
boosted trees	0.309 (1.05x)	0.127 (2.47x)	0.089 (0.45x)

Table A1: Experiment 1. Fitted 2025-08-01, scored 2026-08-01 on 44,316 unseen CVEs (597, 192 and 405 positives; 111, 41 and 70 in training).

No learned aggregation ranks better. On every target, the logistic model's AP is within noise of LEV's:

- union: difference -0.134 to +0.035;
- union without KEV: -0.137 to +0.021;
- CISA KEV: -0.229 to +0.001, and significantly below current EPSS (-0.354 to -0.124).

The boosted model is significantly worse than LEV on all three targets. The rule picked logistic in five of six splits. Peak EPSS alone is as good as anything.

An apparent gain was an artifact. A first version of this experiment used un-normalized EPSS and kept KEV-listed CVEs in the target. There, the logistic model beat LEV (AP 0.435 against 0.397). The gain disappears when either condition is removed:

- with version-normalized inputs, AP is 0.398 against 0.399;
- with KEV members dropped, the difference is -0.051 to +0.006 against LEV, and -0.029 to -0.004 against current EPSS.

A model trained on lists that feed EPSS learns the feedback.

test origin	positives / N	base rate	LEV	current EPSS	logistic	boosted trees
2025-08-02	71 / 32,223	0.22%	0.035	0.042	0.019	0.007
2025-10-31	105 / 43,059	0.24%	0.029	0.027	0.038	0.026
2026-01-30	97 / 49,678	0.20%	0.020	0.025	0.022	0.010

Table A2: Experiment 2, union target. AP for first listing in the next 182 days, among unlisted CVEs at least 30 days old.

Nothing anticipates the lists. Across three forward origins, no method's AP is significantly above current EPSS. The boosted model is significantly below it at two origins, and the logistic model at one. Current EPSS is never significantly worse than LEV, and is slightly better at two of the three origins with CISA and SENSOR dates.

59-75% of the listings in each window concern CVEs first scored less than 30 days before the window, so no history-based method can see them in advance.

Threats.

- Training sets are small (24-111 positives).
- Experiment 1 has one split.
- There are many intervals and no multiplicity correction, so these results are exploratory.
- SENSOR is an EPSS label source, so some circularity remains even without KEV.

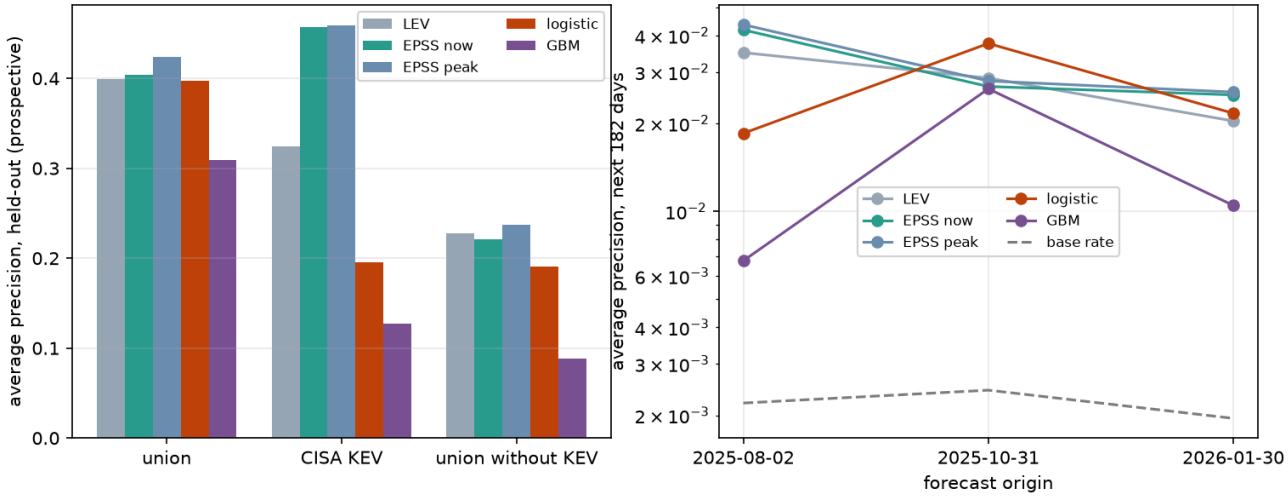


Figure A1. Left: Experiment 1 average precision by target; the learned models never beat LEV or current EPSS. Right: Experiment 2 average precision by forecast origin.

Appendix B: further robustness

- **Reproduction against analysis date** (Section 5.1): `results/validate_dn_grid.json`.
- **Calibration by age and maturity rule** (Table 1a): `results/calibration.json`.
- **Growth families, HAC and block-bootstrap comparisons, and EPSS for never-listed CVEs** (Section 5.4): `results/dn_sweep.json`, Figure B1.
- **Version artifact under percentile, step-only and cohort-fit maps, and the v4 and v5 one-day steps** (Section 5.3): `results/version_normalize.json`.
- **Matched-control early-signal analysis** (Section 5.5): `results/leakage.json`.

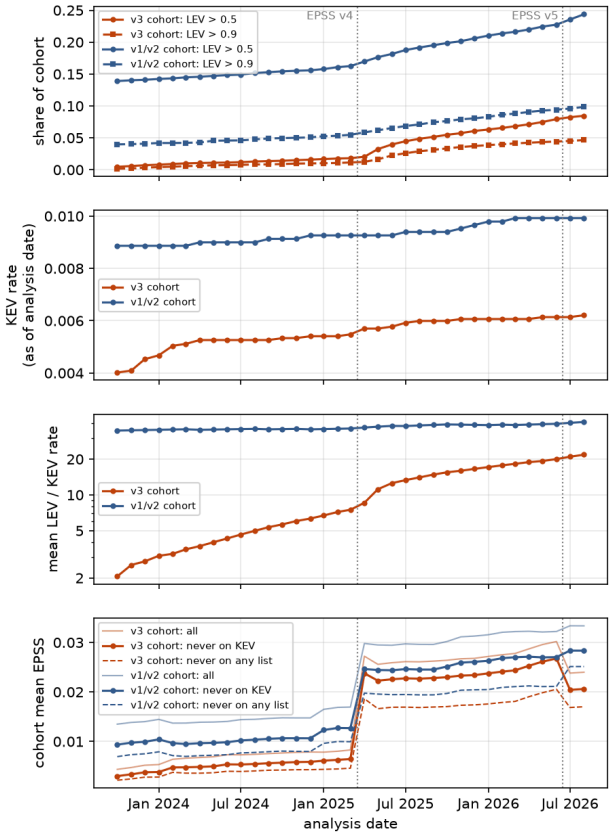


Figure B1. Analysis-date sweep: additional panels, and mean EPSS for CVEs no list records.

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See `references.md` for full entries and verification status.

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